



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY B.Tech.

Course: : Heat Transfer

Course Code: PCC-PE312

Date: 20.06.2025

Time: 10.00 am to 01.00 pm

Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Derive the general three-dimensional heat conduction equation for a cartesian coordinate system using heat conduction processes across six faces. Draw a neat sketch.	10	CO1	BT2
Q.2	A steam main 320 mm in diameter (OD) and 100 m long is covered with 50 mm of high temperature insulation ($k = 0.065 \text{ W/m.K}$) and 40 mm of low temperature as insulation ($k = 0.095 \text{ W/m.K}$). The inner and outer surface temperature as measured are 475°C and 50°C respectively. Calculate: 1) The total heat loss per hour. 2) The total heat loss per m^2 of outer surface. 3) The heat loss per m^2 of the pipe surface. 4) The temperature between the two layers of insulation. Neglect heat conduction resistance through pipe material.	10	CO2	BT4
Q.3	Explain the different modes of heat transfer, and Explain the concept of critical thickness of insulation.	10	CO2	BT2
Q.4	A crucible handle has rectangular cross section of size 5 mm x 2.5 mm. Find the length required so that its end temperature will not exceed 26°C even when the spoon is dipped in boiling water at 110°C for long time. Take K for fin material = 35 W/m.K , $h=11 \text{ W/m}^2\text{K}$, ambient temperature at 20°C . Assume the spoon as infinitely long fin.	10	CO6	BT4
Q.5	Define the Newton's law of cooling. Write its mathematical expression for it, and explain heat transfer coefficient, and its dependence on various factors affecting heat transfer.	10	CO6	BT2
Q.6	What is thermal boundary layer. Explain the concept for flow over a heated flat plate with neat sketch.	10	CO5	BT2
Q.7	Define the following dimensionless terms and write their mathematical formula. (i) Nusselt number (ii) Prandtl number (iii) Reynolds number	10	CO5	BT2

Q.8	Prove that when one radiation shield is inserted between two, nonblack, parallel plates, the heat transfer is reduced to 50% as compared to heat transfer without the shield.	10	CO2	BT1
Q.9	Two opposed, parallel, infinite planes are maintained at 510 K and 460 K respectively. Calculate the heat transfer per unit area between these planes if one plane has an emissivity of 0.8 and the other plane has an emissivity of 0.7.	10	CO3	BT3
Q.10	What is a heat exchanger. Explain Parallel flow and Counter flow heat exchangers with neat sketches for temperature variation across the length of the heat changer.	10	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

- On successful completion of this course, students should be able:
- CO 1 Define, differentiate and interpret basic modes of heat transfer.
 - CO 2 Compute temperature distribution and heat transfer rate for 1-D problems
 - CO 3 Design heat exchanger as per the requirement.
 - CO 4 Select suitable heat exchanger, based on the heat transfer process, geometry and construction for a given application.
 - CO 5 Understand significance of different non dimensional numbers in convective heat transfer.
 - CO 6 Analysis of the extended surface (Fins).



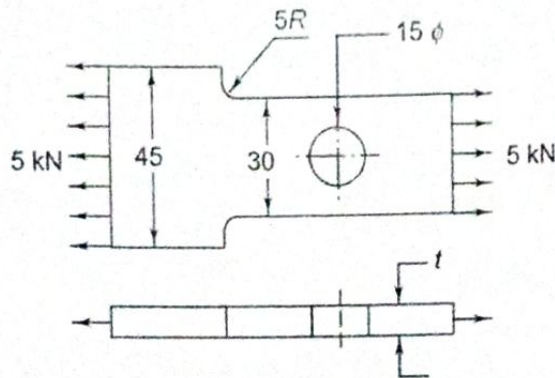
Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY B.Tech.	Course: Machine Design	Course Code: PCC-PE302
Date: 19.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- Attempt all the questions.
- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Use of non-programmable calculator is allowed.

Q.N.		Marks	CO	BT										
	Solve the following questions.													
Q.1	Define Machine Design and Mechanical Engineering Design. Explain the phases and iterations of the design process with the help of flowchart. Explain in detail the classifications of Machine Design.	20	CO1	BT2										
Q.2	Define the following mechanical properties of the metals. <table border="1"> <tr> <td>a. Strength</td><td>b. Stiffness</td><td>c. Elasticity</td><td>d. Plasticity</td><td>e. Ductility</td></tr> <tr> <td>f. Brittleness</td><td>g. Malleability</td><td>h. Toughness</td><td>i. Machinability</td><td>j. Resilience</td></tr> </table>	a. Strength	b. Stiffness	c. Elasticity	d. Plasticity	e. Ductility	f. Brittleness	g. Malleability	h. Toughness	i. Machinability	j. Resilience	20	CO1	BT1 BT3
a. Strength	b. Stiffness	c. Elasticity	d. Plasticity	e. Ductility										
f. Brittleness	g. Malleability	h. Toughness	i. Machinability	j. Resilience										
Q.3	Explain in detail the types of threaded fasteners such as Bolts, Screws and Studs with neat labelled sketches. Describe the different types of thread forms with proper labelled figures.	20	CO2 CO5	BT2 BT6										
Q.4	A flat plate subjected to a tensile force of 5 kN is shown in figure below. The plate material is grey cast iron FG 200 and the factor of safety is 2.5. Determine the thickness of the plate. (Given stress concentration factor $[K_t]$, a. At the fillet section, $[K_t] = 1.8$ b. At the hole section, $[K_t] = 2.16$)	20	CO4	BT4 BT5										



Q.5 The nominal diameter of a triple-threaded square screw is 50 mm, while the pitch is 8 mm. It is used with a collar having an outer diameter of 100 mm and inner diameter as 65 mm. The coefficient of friction at the thread surface as well as at the collar surface can be taken as 0.15. The screw is used to raise a load of 15 kN. Using the uniform wear theory for collar friction, calculate:

- Torque required to raise the load;
- Torque required to lower the load; and
- The force required to raise the load, if applied at a radius of 500 mm.

20

CO3

BT5

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Understand basic design procedure and the necessary design considerations.
CO2	Define and distinguish static and fluctuating loads/stresses.
CO3	Derive empirical relations for different machine elements using relevant theories of failure.
CO4	Analyse behaviour of machine components under static and fluctuating loads/stresses.
CO5	Design shafts, gears, clutches, power screws etc. for real life applications when subjected to static loading.